

Sprint 3 Schedule Analysis

Network Dependencies & Critical Path

📅 November 29, 2025

⚠️ Executive Summary

Critical Path: A → B → D → E (13 days total)

Sprint Risk Level: High

Key Success Factor: Club infrastructure must be completed before profiles can be built

1 🧩 Task Dependencies

Our Sprint 3 consists of 15 tasks organized into five distinct work streams. Below is the complete breakdown of task dependencies:

1.1 🏠 Club Infrastructure Path (CRITICAL PATH)

- **Task A:** Club Creation Backend
→ No dependencies — Duration: 3 days
 - **Task B:** Club Editing & Deletion Backend
→ Depends on Task A — Duration: 2 days
 - **Task C:** Club Creation/Editing UI
→ Depends on Task B — Duration: 3 days
 - **Task D:** Event & Club Banner & ID Defaults
→ Depends on Task A — Duration: 2 days
 - **Task E:** Club Profiles Frontend
→ Depends on Task D — Duration: 3 days
- ⚡ **This forms the critical path: A → B → D → E (13 days total)**

1.2 🛡️ Security Enhancement Path

- **Task F:** Security Fixes (Token-based endpoint protection)
→ No dependencies — Duration: 3 days
- 📌 *Note: This is foundational for all authenticated operations but runs in parallel.*

1.3 Event Management Path

- **Task G:** Event Editing Backend
→ Depends on Task F — Duration: 2 days
- **Task H:** Event Deletion Backend
→ Depends on Task F — Duration: 1 day
- **Task I:** Event Editing/Deletion UI
→ Depends on Tasks G AND H — Duration: 2 days

1.4 Reviews & Engagement Path

- **Task J:** Reviews/Comments Backend
→ Depends on Task F — Duration: 3 days
- **Task K:** Reviews/Comments Frontend
→ Depends on Task J — Duration: 2 days

1.5 Search & Calendar Integration Path

- **Task L:** Search Events Backend (Direct matches)
→ No dependencies — Duration: 2 days
- **Task M:** Search Events Frontend
→ Depends on Task L — Duration: 2 days
- **Task N:** Calendar Endpoint Backend
→ No dependencies — Duration: 2 days
- **Task O:** Connect Calendar Endpoint to Calendar Page
→ Depends on Task N — Duration: 1 day

1.6 Aggregator Path

- **Task P:** Post Event (Aggregator)
→ No dependencies — Duration: 2 days
-  *Note: Independent scraper integration work.*

2 Network Diagram

The network diagram below would illustrate the task dependencies and highlight the critical path in red.

Sprint 3 Network Dependency Diagram

Critical Path: A → B → D → E (10 days)

Critical Path

Security

Event Management

Reviews

Search & Calendar

Aggregator

Figure 1: Title img

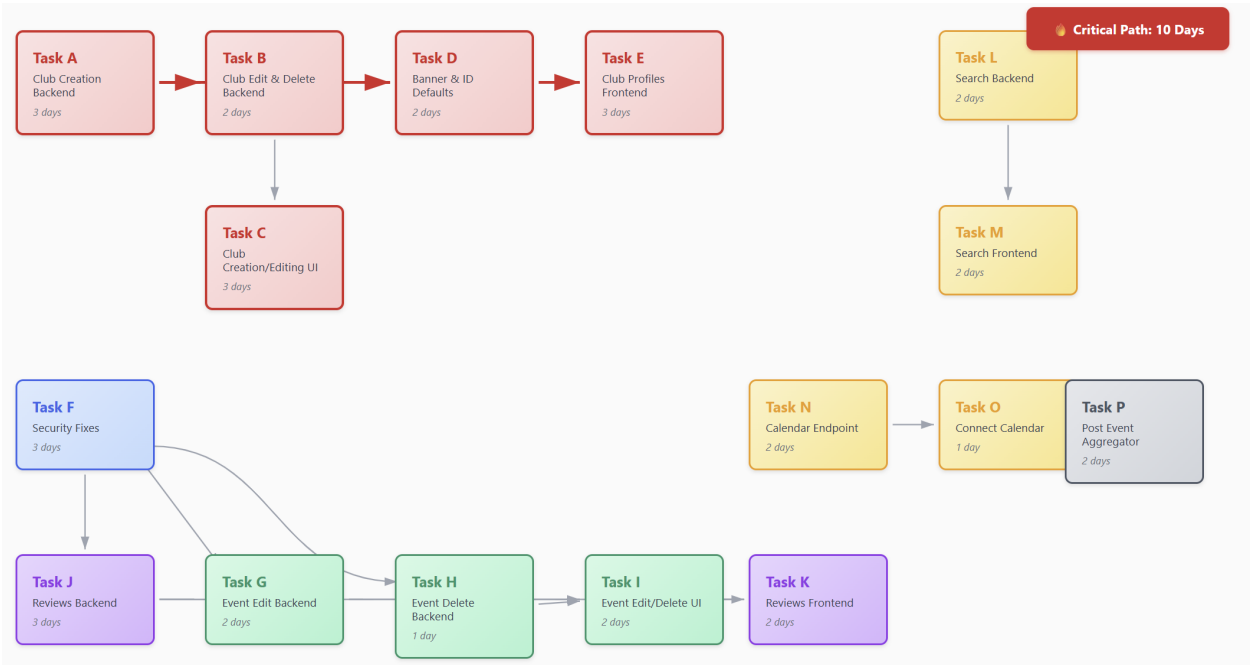


Figure 2: html img

3 🧩 Critical Path Analysis

3.1 Identified Critical Path

A → B → D → E

Task A: Club Creation Backend (3 days)

Task B: Club Editing & Deletion Backend (2 days)

Task D: Event & Club Banner & ID Defaults (2 days)

Task E: Club Profiles Frontend (3 days)

Total: **10 days**

Note: Path A → B → C also totals 8 days but

A → B → D → E is longer at 10 days

This represents the longest dependency chain in the project and determines the minimum time required to complete the sprint. **Any delay in these tasks will directly impact the sprint completion date.**

3.2 Why This is Critical

The club infrastructure represents foundational functionality that blocks multiple features:

1. 🏠 Club profiles cannot exist without the club creation system (Task A)
2. Club editing and deletion must be tested before profiles go live (Task B)
3. 🖼️ Banner and ID systems are required for proper club display (Task D)
4. 👤 Club profile pages with executive information depend on all prior infrastructure (Task E)
5. 📅 This forms the longest linear dependency chain in the project at 10 days

4 ✅ Keeping the Sprint on Schedule

To ensure our sprint stays on schedule, we implement the following strategies:

4.1 🧭 Early Start on Critical Path Tasks

- Begin Task A on **Day 1** of the sprint
- Assign most experienced backend developers to Tasks A and B
- Prioritize code reviews and testing for Tasks A, B, D, and E
- Hold daily standups specifically focused on critical path progress
- Ensure Task D starts immediately after A completes (parallel to B)

4.2 ☰ Parallel Work Streams

- Security fixes (Task F) can start Day 1 and enable event management path
- Search functionality (L → M) is completely independent
- Calendar integration (N → O) runs in parallel
- Aggregator work (Task P) is isolated
- This allows 5+ teams to work simultaneously without blocking each other

4.3 Risk Mitigation

- 🕒 **Buffer time:** Build 1-2 day buffer into critical path estimates
- 🔗 **Early integration:** Test A + B integration as soon as B completes
- 🗄️ **Database schema first:** Define club schema and banner system before Day 1
- 💾 **Mock data:** Frontend teams can work with mock club data while backend progresses
- 👤 **Executive profiles:** Plan user linking architecture upfront to avoid rework in Task E
- 🛡️ **Security dependency:** Ensure Task F completes early to unblock G, H, and J

4.4 📊 Progress Monitoring

- Track critical path task completion daily
- Use burn-down charts specifically for A → B → D → E
- Monitor Task F closely as it gates three other features (G, H, J)
- Identify blockers within 4 hours of occurrence
- Escalate any critical path delays to the entire team immediately

5 ⚠️ What Could Go Wrong: Post-Sprint Analysis

5.1 Most Likely Failure Scenarios

If we were unable to finish the sprint on time, the following scenarios (illustrated in Figure ??) would be the most probable causes:

5.1.1 🚩 Scenario 1: Club Creation Complexity (Task A)

What went wrong: Task A took 5 days instead of 3 days

🔍 Root cause: Database schema for clubs more complex than anticipated; relationships between clubs, executives, and events required additional modeling

! Cascade effect: Tasks B, D, and E are all blocked, pushing sprint completion by 2 days minimum

💡 **What we learned:** Should have prototyped club schema during Sprint 2. Need database design review before sprint start for major new entities.

5.1.2 🚩 Scenario 2: Banner & ID System Underestimation (Task D)

What went wrong: Banner upload, image processing, and default ID generation took 4 days instead of 2

🔍 **Root cause:** File upload infrastructure, image resizing, storage integration, and unique ID generation logic more complex than estimated

⚠️ **Cascade effect:** Task E (Club Profiles) blocked, extending sprint by 2 days

💡 **What we learned:** Media handling features always take longer than expected. Should have used existing image upload libraries (Cloudinary, AWS S3) instead of custom solutions. Need to spike on file handling before estimation.

5.1.3 🚩 Scenario 3: Executive Profile Linking Issues (Task E)

What went wrong: Linking executives to their user profiles and displaying contact information proved challenging

🔍 **Root cause:** User permission system wasn't robust enough; privacy concerns about displaying executive contact info; UI complexity for showing club hierarchies

⚠️ **Cascade effect:** Club profiles incomplete at sprint end, core feature not delivered

💡 **What we learned:** Should have clarified privacy and permission requirements during planning. Need clear user stories for "who can see what" before building profile features.

5.2 Secondary Risk: Security Bottleneck (Task F)

Task F Dependency Risk

While Task F is not on the critical path, it gates three important features:

- Task G (Event Editing) requires token authentication to verify club ownership
- Task H (Event Deletion) requires same authentication system
- Task J (Reviews/Comments) needs token-based edit/delete permissions
- If F is delayed, these three features cannot begin, potentially cutting scope
- **Mitigation:** Start F on Day 1 alongside A; assign security-experienced developer; use existing JWT infrastructure from Sprint 2

5.3 Tertiary Risk: Integration Complexity

Multiple Integration Points

Sprint 3 has several integration challenges:

- Club profiles (Task E) must integrate with existing event system
- Reviews (Tasks J, K) integrate with both events and clubs
- Search (Tasks L, M) must query across events and clubs
- Calendar integration (Tasks N, O) connects to existing calendar view
- **! Risk:** Integration points often reveal edge cases and data inconsistencies
- **Mitigation:** Define clear API contracts; test integrations early; maintain consistent data models across features

6 Key Learnings & Recommendations

6.1 Architectural Insights

1. **Club infrastructure is foundational:** Just like event feed in Sprint 2, clubs are a core entity that everything else builds upon. This deserves top priority.
2. **Security as an enabler:** Task F gates multiple features. Getting security right early unblocks significant work.
3. **Independent features provide flexibility:** Search, calendar integration, and aggregator can progress regardless of critical path status, allowing for resource reallocation.
4. **Media handling needs planning:** Banner and image features (Task D) require upfront architecture decisions about storage, processing, and delivery.

6.2 Process Improvements

1. **Database schema design sessions:** Hold pre-sprint meetings to design club schema, relationships, and constraints before Day 1
2. **Dual critical path monitoring:** Watch both $A \rightarrow B \rightarrow D \rightarrow E$ and Task F, as F gates multiple features
3. **Media strategy upfront:** Decide on image storage solution, upload libraries, and processing pipeline before sprint
4. **Privacy and permissions planning:** Document who can see/edit what before building club profile and review features
5. **Integration testing early:** Test club-event integration, review-authentication integration, and search-database integration as soon as backend tasks complete
6. **Buffer allocation:** Add 20% buffer to Task A (3 days $\rightarrow$ 4 days) given its foundational importance

6.3 </> Technical Recommendations

1. Use established image upload services (Cloudinary, imgix, AWS S3) rather than custom solutions
2. Implement role-based access control (RBAC) properly in Task F to support club executives, moderators, and users
3. Create reusable club profile components that can be used in search results and event pages
4. Use full-text search capabilities (PostgreSQL full-text, Elasticsearch) for Task L rather than simple SQL LIKE queries
5. Build API versioning strategy now to handle calendar endpoint changes gracefully
6. Document all club-related database schemas with clear foreign key relationships
7. Implement proper cascade delete behavior for clubs and their associated events/reviews

7 🚩 Conclusion

The critical path analysis reveals that Sprint 3 success depends heavily on the **club infrastructure path** ($A \rightarrow B \rightarrow D \rightarrow E$). Additionally, Task F acts as a gating factor for event management and reviews. By identifying these dependencies early, we can:

- ✔ Start foundational work (A and F) on Day 1
- ✔ Allocate senior developers to critical path
- ✔ Enable parallel work on search, calendar, and aggregator
- ✔ Plan for media handling and privacy concerns upfront
- ✔ Monitor dual dependencies (critical path + security)
- ✔ Test integrations early and often

Sprint 3 introduces the club entity as a first-class citizen in our system. This is comparable to how Sprint 2 established the event feed. The difference is that clubs have more complex relationships (executives, banners, events, reviews) which increases integration risk. The network diagram clearly shows that delays in club creation will cascade through the sprint, while delays in search or calendar features can be absorbed. This understanding allows us to make informed decisions about task prioritization and resource allocation throughout Sprint 3.